

■ 1.1.5 Advanced Topic: Complex Numbers

You can enter complex numbers in *Mathematica* just by including the constant I , equal to $\sqrt{-1}$. Make sure you type a capital I .

This gives the imaginary number result $2i$.

```
In[1]:= Sqrt[-4]
Out[1]= 2 I
```

This gives the ratio of two complex numbers.

```
In[2]:= (4 + 3 I) / (2 - I)
Out[2]= 1 + 2 I
```

Here is the numerical value of a complex exponential.

```
In[3]:= Exp[2 + 9 I] //N
Out[3]= -6.73239 + 3.04517 I
```

$x + I y$	the complex number $x + i y$
$\text{Re}[z]$	real part
$\text{Im}[z]$	imaginary part
$\text{Conjugate}[z]$	complex conjugate z^* or $\bar{z}$
$\text{Abs}[z]$	absolute value $ z $
$\text{Arg}[z]$	the argument ϕ in $ z e^{i\phi}$

Complex number operations.